EV Adoption Analytics: Washington State Electric Vehicle Population Intelligence

Section C | Group 6 | Submission: April 2026

1. Cover Page

Field	Details
Project Title	EV Adoption Analytics: Washington State Electric Vehicle Population Intelligence
Sector	Transportation & Energy Policy
Team ID	Section-C_G-6
Institute	Newton School of Technology
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Submission Date	April 2026
Tableau Workbook	`EV_Adoption_Analytics.twbx`
Processed Dataset	`data/processed/EV_Final_Ready_Tableau.csv`

Team Members

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2. Executive Summary

Problem

Washington State is the leading EV market in the USA, yet policymakers, utility providers, and automakers lack a unified analytical view of where adoption is happening, which vehicle technologies are winning, and which infrastructure bottlenecks need immediate attention.

Approach

We sourced the complete Washington State EV registration registry (280,833 raw records) from the Department of Licensing, performed a rigorous 5-stage ETL pipeline, engineered 9 derived analytical features, and built three interactive Tableau dashboards covering executive KPIs, geospatial intelligence, and market & technology analysis.

Key Insights

1. **BEV technology is winning** — BEVs account for 68.7% of verified-range vehicles, up sharply from near-zero pre-2013.
2. **Tesla dominates at 23.9%** of all registrations — more than double its nearest competitor.
3. **King County holds 46% of all EVs** — massive geographic concentration in the Seattle metro.
4. **82.1% of verified-range vehicles are CAFV-eligible**, validating the policy incentive structure.
5. **Luxury segment at 48.1%** signals that EVs remain predominantly an upper-income adoption story.
6. **Infrastructure friction is highest in rural counties** (Asotin, Mason, Jefferson avg score 3.0) despite low EV density.
7. **Battery range has improved 6× since 2010** — from ~30 mi to ~200 mi average for long-range BEVs.
8. **Mass Adoption wave (2021+) is accelerating** — 37,530 vehicles in just 4–5 model years vs 50,005 across the full 5-year Growth wave.

Key Recommendations

1. Prioritise charging infrastructure investment in Snohomish, Pierce, and Clark counties — highest growth outside King.

2. Extend CAFV incentives to mass-market PHEVs to convert the 17.9% ineligible segment.
3. Target grid upgrade partnerships in high-friction counties (Asotin, Mason, Jefferson) before EV demand arrives.
4. Design affordable BEV incentive programs to shift the 51.9% mass-market share toward full electrification.
5. Use the Tesla Model 3 and LEAF as benchmark adoption signals for regional demand forecasting.

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3. Sector and Business Context

Sector Overview

The global EV transition is the largest structural shift in automotive history. Washington State leads US adoption driven by progressive clean-energy legislation (Clean Cars 2030 Act), state EV tax credits, and a high-income, tech-forward population concentrated in the Puget Sound corridor. The state mandates all new passenger car sales be zero-emission by 2035.

Decision-Makers / Stakeholders

Stakeholder	Decision Supported
WA State Office of Clean Energy	Infrastructure investment prioritisation
Electric Utility Providers	Grid upgrade planning by county
Auto Manufacturers	Market entry strategy, model mix decisions
City & County Planners	Charging station siting and zoning
Policy Analysts	Incentive program design and evaluation

Why This Problem Matters

Without granular, spatially-aware analytics, infrastructure investment risks being misallocated — either over-serving already-saturated urban counties or failing to prepare rural areas for imminent demand surges. Every year of delayed grid investment in high-friction counties adds disproportionate coordination cost when EV density eventually arrives.

Core Business Question

> *Which counties, vehicle segments, and technology tiers should Washington State prioritise for infrastructure investment and incentive design to maximise equitable EV adoption by 2030?*

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4. Problem Statement and Objectives

Formal Problem Definition

Washington State possesses detailed EV registration data but lacks a decision-ready analytical framework that simultaneously answers: **Where** are EVs concentrated? **Which** technologies and brands are leading? **Who** is adopting — luxury or mass-market buyers? **Where** is grid infrastructure under-prepared?

Scope

- Geography: Washington State (all 39 counties, 500+ cities)
- Time period: Model Year 1999 – 2026
- Dataset: 280,833 raw registration records → 101,275 verified-range analytical records
- Technology: Python ETL + Tableau Public dashboards

Success Criteria

- Deliver 3 interactive Tableau dashboards answering geographic, executive, and market dimensions
- Produce 8–12 data-backed insights in decision language
- Generate 3–5 actionable recommendations with measurable expected impact
- Complete full ETL pipeline with documented data dictionary and cleaning log

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5. Data Description

Source

Attribute	Detail
Name	Electric Vehicle Population Data
Provider	Washington State Department of Licensing
Access	https://catalog.data.gov/dataset/electric-vehicle-population-data
Format	CSV, UTF-8, comma-delimited

Attribute	Detail
Raw Row Count	280,833
Analytical Row Count	101,275 (verified electric range only)
Columns	26 (17 original + 9 engineered)
Granularity	One row = one registered electric vehicle
Time Span	Model Year 1999 – 2026

Key Original Columns

Column	Type	Description
`County`	String	WA county of registration (170 found, 39 standard)
`City`	String	City of registration
`Model Year`	Integer	Manufacture year (1999–2026)
`Make`	String	Vehicle manufacturer (36 unique brands)
`Model`	String	Vehicle model name
`Electric Vehicle Type`	String	BEV or PHEV
`Electric Range`	Float	EPA electric-only range in miles
`CAFV Eligibility`	String	Clean Alternative Fuel Vehicle program status
`Electric Utility`	String	Utility provider(s), `||`-separated for multi-jurisdiction
`Vehicle Location`	String	`POINT(lon lat)` coordinate string
`Legislative District`	Float	WA State legislative district (1–49)

Data Quality Issues Found

Issue	Scale	Resolution
Electric Range = 0 (“not researched”)	~64% of raw rows (179,558)	Replaced with NaN; retained in `Electric Range Raw` for population counts
Postal Code stored as float (98034.0)	All rows	Converted to integer string
Missing County / City / Legislative District	Minor (<1%)	Filled with `"Unknown"` to preserve grouping integrity
Corrupt POINT coordinate strings	~500 rows	Dropped — required for map integrity

Issue	Scale	Resolution
CAFV eligibility in verbose raw text	All rows	Simplified to 3-tier `Eligibility_Status`

6. Cleaning and Transformation

Step 1 — Initial Setup & Data Loading

Libraries used: `pandas`, `numpy`

The raw dataset `Electric_Vehicle_Population_Data.csv` was loaded into a pandas DataFrame. A `DtypeWarning` for Column 11 (`Electric Range`) was noted due to mixed types — handled implicitly by pandas during read.

Item	Detail
Raw file loaded	`Electric_Vehicle_Population_Data.csv`
Initial shape	(280,813 rows × 17 columns)
Warning noted	DtypeWarning on Column 11 — mixed types in Electric Range

Step 2 — Missing Geographic Data

Missing values in the following geographic columns were filled with the string `'Unknown'` to prevent row loss during grouping and aggregation:

Column Filled	Reason
`County`	Required for county-level grouping and choropleth map
`City`	Required for city-level drill-down filter
`Postal Code`	Required for ZIP-level mapping
`Legislative District`	Required for policy-level analysis
`Vehicle Location`	Required for coordinate extraction

Step 3 — Postal Code Standardisation

The `Postal Code` column was originally stored as a float (e.g., `98034.0`) due to how pandas inferred types from CSV. It was standardised by:

1. Converting to string type
2. Stripping the trailing `.0` suffix (e.g., `98034.0` → `98034`)

This ensures correct ZIP-level mapping in Tableau without decimal artifacts in labels.

Step 4 — Handling Electric Range

`Electric Range Raw` **column created** — an exact copy of the original `Electric Range` column was preserved as `Electric Range Raw` before any transformation. This retains the original values (including zeros) for population count analyses.

Zero value treatment — All zero values in the primary `Electric Range` column were replaced with `np.nan`. This is because a `0` value in the DOL registry indicates **"range not researched"**, not a literal zero-mile vehicle. Using these zeros in statistical analyses would artificially deflate the average range and skew distributions.

Before Treatment	After Treatment
179,538 rows with `Electric Range = 0`	Replaced with `NaN`
Mean range (incl. zeros)	Meaningless / deflated
Mean range (NaN-excluded)	**107.5 mi** — accurate fleet average

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Step 5 — Feature Engineering (KPI Preparation)

5a. Vehicle Age

- Range: 0 (2026 model) to 27 (1999 model)
- Purpose: Fleet aging analysis, age-vs-range correlation

5b. CAFV_Status

The verbose `Clean Alternative Fuel Vehicle (CAFV) Eligibility` column was simplified:

Value	Count
`Eligible`	76,897
`Unknown/Not Eligible`	24,074

Step 6 — Advanced Feature Engineering for Tableau

6a. Range_Category

A categorical column derived from `Electric Range Raw` using predefined bins:

Bin	Label	Count
`[-1, 0]`	`Unknown/Not Researched`	179,538 (raw data only)
`(0, 100]`	`Urban Commuter`	63,848

Bin	Label	Count
`(100, 250]`	`Regional Traveler`	27,529
`(250, 1000]`	`Long Range`	9,594

6b. Geospatial — Latitude & Longitude

A helper function `extract_coords` was defined using **regular expressions** to parse the `Vehicle Location` POINT string:

```
# Example input:  POINT (-122.22901 47.72201)
# Outputs: Longitude = -122.22901, Latitude = 47.72201
```

Two new numeric columns `Latitude` and `Longitude` were created, enabling geographical scatter maps and heatmaps in Tableau.

6c. Market_Segment — Brand Segmentation

Segment	Count	Share
`Mass Market`	64,595	64%
`Luxury`	36,376	36%

6d. Eligibility_Status — Traffic Light Classification

The CAFV eligibility field was mapped into a clean 3-tier status:

Input Text Contains	Assigned Status
`"Eligible"`	`Eligible`
`"not eligible"` / low range	`Not Eligible`
`"unknown"` / not researched	`Unknown`

Final distribution: **Eligible (76,897)** · **Not Eligible (24,074)**

6e. Utility_Complexity_Score — Infrastructure Friction

- Score of `1` = single utility (e.g., `PUGET SOUND ENERGY INC`)
- Score of `2` = two utilities (e.g., `PUGET SOUND ENERGY INC||CITY OF TACOMA`)
- Score of `3` = three or more utility jurisdictions
- Higher score = more administrative coordination required for grid upgrades

Step 7 — Dropping Rows with Missing Electric Range

Decision rationale: 179,538 out of 280,813 rows had `NaN` in `Electric Range Raw` — meaning the DOL had not researched the range for these vehicles. For range-specific KPIs and analyses, including these records would produce meaningless averages.

Action: All rows where `Electric Range Raw` is `NaN` were dropped.

Stage	Shape
Before drop	(280,813 × 26)
After drop	**(101,275 × 26)**
Rows retained	101,275 (~36.06% of original)

This analytical subset is the dataset used for all KPIs, statistical tests, and Tableau dashboards.

Step 8 — Final Cleanup & Export

Geospatial filtering: Rows where `Latitude` or `Longitude` could not be extracted (corrupt or null POINT strings) were removed to ensure map integrity.

Final export:

Property	Value
Output filename	`EV_Final_Ready_Tableau.csv`
Final shape	101,275 rows × 26 columns
Encoding	UTF-8
Also exported	`EV_Population_Cleaned.csv` (intermediate cleaned file)

Summary: All 9 Engineered Columns

#	Column	Source	Logic	Purpose
1	`Electric Range Raw`	`Electric Range`	Copy before NaN treatment	Preserve original values for population counts
2	`Vehicle Age`	`Model Year`	`2026 - Model Year`	Fleet aging analysis
3	`CAFV_Status`	`CAFV Eligibility`	Contains 'Eligible' → 'Eligible' else	Simplified incentive KPI

#	Column	Source	Logic	Purpose
			'Unknown/Not Eligible'	
4	`Range_Category`	`Electric Range Raw`	Bins: 0–100 / 101–250 / 251+	Human-readable range tier
5	`Latitude`	`Vehicle Location`	Regex extraction from POINT string	Geospatial mapping
6	`Longitude`	`Vehicle Location`	Regex extraction from POINT string	Geospatial mapping
7	`Market_Segment`	`Make`	Luxury brand list → 'Luxury'; else 'Mass Market'	Socio-economic accessibility
8	`Eligibility_Status`	`CAFV Eligibility`	3-tier mapping: Eligible / Not Eligible / Unknown	Dashboard traffic-light filter
9	`Adoption_Wave`	`Model Year`	≤2015 Early; 2016–2020 Growth; 2021+ Mass Adoption	Transition velocity analysis
10	`Utility_Complexity_Score`	`Electric Utility`	Count of `|`-split segments	Infrastructure friction proxy

7. KPI Framework

KPI	Definition	Formula	Value (Verified Range Dataset)
Total Registrations	All verified-range EV records	`COUNT(DOL Vehicle ID)`	**101,275**
BEV Adoption %	Share of Battery Electric Vehicles	`COUNT(BEV) / COUNT(All)`	**68.7%** (Dashboard 1 filtered view)
Avg Electric Range	Mean EPA range across all vehicles	`AVG(Electric Range)`	**151.7 mi**

KPI	Definition	Formula	Value (Verified Range Dataset)
CAFV Eligibility %	Share of CAFV-eligible vehicles	`COUNT(Eligible) / COUNT(All)`	**82.1%**
Luxury Segment %	Share of luxury-brand EVs	`COUNT(Luxury) / COUNT(All)`	**48.1%** (filtered view)
Top County Share	King County as % of total	`COUNT(King) / COUNT(All)`	**46.0%**
Tesla Market Share	Tesla as % of all makes	`COUNT(Tesla) / COUNT(All)`	**23.9%**
Mass Adoption Count	Vehicles from 2021+ Model Year	`COUNT(Adoption_Wave = 'Mass Adoption')`	**37,530**
Avg Utility Complexity	Mean utility jurisdiction score	`AVG(Utility_Complexity_Score)`	**1.8**

8. Exploratory Analysis

Major Trends

Adoption Curve (1999–2026)

EV registrations in Washington State followed a classic S-curve. Growth was negligible pre-2010 (< 200 vehicles/year). The 2013–2016 period marked inflection — Nissan LEAF democratised BEV ownership. The 2018–2019 Tesla Model 3 launch triggered exponential acceleration. The 2020–2021 dip reflects supply chain disruptions; 2022–2026 shows resumed rapid growth with mass-market PHEVs entering at scale.

Adoption Wave	Model Years	Count	Share
Early Adopters	≤ 2015	13,436	13%
Growth Phase	2016 – 2020	50,005	49%
Mass Adoption	2021+	37,530	37%

Geographic Concentration

The top 5 counties account for 76% of all registrations — a highly concentrated urban adoption pattern.

County	Registrations	Share
King	46,605	46.0%
Snohomish	10,718	10.6%

County	Registrations	Share
Pierce	8,590	8.5%
Clark	6,814	6.7%
Thurston	4,103	4.1%

Top Cities: Seattle (16,230) → Vancouver (4,254) → Bellevue (3,827) → Olympia (2,668) → Redmond (2,645)

Manufacturer Landscape

Make	Count	EV Type Focus
TESLA	24,182	BEV only
CHEVROLET	9,848	Mixed (BOLT EV + VOLT PHEV)
TOYOTA	9,750	PHEV dominant (Prius Prime, RAV4 Prime)
NISSAN	9,631	BEV dominant (LEAF)
BMW	7,048	Mixed (i3 BEV + PHEV models)
JEEP	6,568	PHEV dominant (Wrangler 4xe)

Vehicle Type Split
- BEV: 45,583 (45% of full dataset; 68.7% in verified-range filtered view)
- PHEV: 55,692 (55% of full dataset; 31.3% in verified-range filtered view)

The discrepancy arises because PHEVs disproportionately lack researched range data in the DOL registry.

Market Segment
- Mass Market: 64,595 (64%)
- Luxury: 36,376 (36%)

Range Categories

Category	Miles	Count	Share
Urban Commuter	≤ 50 mi	63,848	63%
Regional Traveler	51–200 mi	27,529	27%
Long Range	> 200 mi	9,594	9%

Segment-Level Insights

- Luxury BEVs average **256 mi** range vs Mass Market BEVs at **98 mi**
- Tesla Model 3 alone = **~14,000** registrations — single largest model

- Chevrolet BOLT EV is the top mass-market BEV at **~6,000** units
- Nissan LEAF remains the most affordable long-term adoption vehicle
- King County's avg utility complexity = **2.0** (multi-agency grid already in effect)

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9. Statistical Analysis

Methods Applied

1. Mann-Whitney U Test — BEV vs PHEV Range
- H_0: No difference in electric range distribution between BEV and PHEV
- Result: $U = 4.2 \times 10^8$, **$p < 0.001$** → Reject H_0
- Interpretation: BEVs have statistically significantly higher electric range. PHEVs cluster at 20–50 mi (Urban Commuter tier); BEVs cluster at 100–330 mi.

2. One-Way ANOVA — Range across Adoption Waves
- H_0: Mean electric range is equal across Early, Growth, and Mass Adoption waves
- Result: $F(2, 101272)$ = **847.3**, **$p < 0.001$** → Reject H_0
- Post-hoc (Tukey): All 3 wave pairs are significantly different
- Interpretation: Each generation of EVs delivers meaningfully higher range — technology progress is statistically confirmed.

3. Pearson Correlation — Vehicle Age vs Electric Range
- r = **−0.61**, $p < 0.001$
- Interpretation: Moderate-strong negative correlation. Older vehicles have substantially shorter range — each additional year of age corresponds to ~8–10 mi lower range on average.

4. Kruskal-Wallis Test — Range by Market Segment
- H_0: Range distribution identical for Luxury and Mass Market
- Result: **$p < 0.001$** → Reject H_0
- Interpretation: Luxury vehicles have a statistically higher range distribution, confirming that premium pricing partially reflects superior battery technology.

5. Infrastructure Complexity Distribution
- Mean Utility Complexity Score: **1.8**
- Counties with Score 3.0: Asotin, Mason, Jefferson — all rural, low-EV-density areas
- King County (highest EV density): Score **2.0** — manageable multi-agency coordination already established

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10. Dashboard Walkthrough

Dashboard 1 — EV Adoption Pulse (Executive Overview)

Objective: Provide policymakers and executives a single-screen command view of WA State EV adoption.

KPI Banner (5 cards across top)

KPI Card	Displayed Value	Color
Total Vehicles	40,661 (filtered)	Electric Blue
BEV Adoption %	68.7%	Electric Blue
Avg Electric Range	151.7 mi	Neon Green
CAFV Eligibility %	82.1%	Purple
Luxury Segment %	48.1%	Amber

Charts
- **EV Registrations by Year** (stacked area): Electric Blue (BEV) + Amber (PHEV) — shows the dramatic growth curve from 2013 onwards with Market Segment (Luxury = purple, Mass Market = green) overlaid
- **Top 10 EV Manufacturers** (horizontal bar): Tesla leads at 16,830 in filtered view; sequential blue gradient
- **Vehicle Type Split** (donut): 68.7% BEV (blue) / 31.3% PHEV (amber)
- **Adoption Waves × Market Segment** (stacked bar): Early (5,910) → Growth (16,909 Luxury + 13,140 Mass) → Mass Adoption

Filters: Make dropdown, Electric Vehicle Type toggle, Market Segment, Year range slider

Aesthetic: Dark navy background (#0A0E1A), electric blue accents, white titles, neon green highlights

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Dashboard 2 — Geographic Intelligence (Regional Map)

Objective: Identify spatial concentration, regional infrastructure needs, and grid complexity patterns.

Charts
- **EV Count by County** (bubble/scatter map on dark basemap): Bubble size = registration count; King County bubble dominates Seattle metro; sparse coverage east of Cascades
- **City-Level EV Distribution** (dot density map): High-precision lat/long scatter; Seattle/Bellevue/Redmond cluster visually prominent; Spokane as the only significant eastern WA cluster
- **Top 10 Counties — BEV vs PHEV** (stacked bar): King (27,486 shown in filtered view) leads massively; BEV (blue) vs PHEV (amber) split visible per county

- **Infrastructure Friction — Avg Utility Complexity** (horizontal bar): Asotin, Mason, Jefferson score 3.0 (highest friction); Clark, King, Cowlitz at 2.0

Key Geographic Insight Visible: The utility complexity chart reveals that rural counties with the fewest EVs today have the most complex utility jurisdictions — making future grid upgrades administratively harder precisely where demand will grow next.

Filters: County dropdown, Electric Vehicle Type, Limit (Top 15 by complexity), interactive map click-to-filter

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Dashboard 3 — Market & Technology Deep Dive

Objective: Analyse competitive brand dynamics, battery technology progress, and incentive program effectiveness.

Charts
- **Brand Competitiveness — BEV vs PHEV** (grouped horizontal bar): Tesla, Nissan, Chevrolet lead BEV; Toyota, BMW, Jeep, Volvo, Chrysler lead PHEV — reveals clear brand technology positioning
- **Battery Technology Evolution — Avg Range by Year** (dual line): Purple = Luxury, Green = Mass Market; both trend upward; "Sweet Spot" reference band at 100–200 mi; Luxury consistently above
- **CAFV Eligibility Status** (pie): 82.06% Eligible (green), 17.94% Not Eligible (red)
- **Range Capability Evolution by Year** (stacked area): Urban Commuter (gray), Regional Traveler (blue), Long Range (green) — Long Range share visibly expanding post-2017
- **Model Mix — Within Top Brands** (treemap): Tesla Model 3 = largest tile; LEAF, BOLT EV, Prius Prime, Wrangler, Pacifica visible as significant models
- **Top Models Table**: BMW X5 (595 count, 20.9 mi avg range) through Nissan LEAF (6,253 count, 107.5 mi avg) — performance vs volume trade-off visible

Filters: Market Segment toggle (Luxury/Mass Market), Year of Model Year slider (2000–2021), Make dropdown, Range Category

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11. Key Insights

1. **BEV technology has decisively overtaken PHEV in verified-range performance** — BEVs average 189 mi vs PHEVs at 32 mi. PHEVs serve a transition role but are not the end-state technology.
2. **Tesla's 23.9% market dominance is structurally unchallenged** — no other brand exceeds 10%. Tesla's Model 3 alone (≈14K units) exceeds all Chevrolet EV models combined.

3. **King County (Seattle metro) absorbs 46% of all EVs** — any infrastructure investment outside this cluster is currently under-utilised relative to need. Snohomish and Pierce represent the highest-ROI next investment zones.
4. **82.1% CAFV eligibility rate validates the current incentive design** for BEVs, but the 17.9% ineligible segment (predominantly short-range PHEVs) represents a policy gap — these buyers purchased EVs but don't qualify for the program.
5. **Luxury brands account for 36–48% of registrations** (depending on filter), revealing that EV adoption in WA State remains significantly skewed toward high-income buyers. Mass-market electrification is the unfinished agenda.
6. **Battery range has improved 6× since 2010** — from a fleet average of ~30 mi to ~180 mi. This validates continued investment in BEV technology and signals that range anxiety is an increasingly obsolete barrier.
7. **The Mass Adoption wave (2021+) is outpacing the Growth wave** on a per-year basis — 37,530 vehicles across 4–5 model years vs 50,005 across 5 model years. The tipping point has been crossed.
8. **Rural counties (Asotin, Mason, Jefferson) have the highest utility complexity scores (3.0)** despite having the fewest EVs. These areas require proactive multi-agency grid coordination agreements before EV demand arrives.
9. **Nissan LEAF is the most accessible mass-market BEV** — 9,631 registrations at a consistent price point; a bellwether for price-sensitive EV adoption and a signal that incentive design should target the $25K–$35K segment.
10. **Vehicle age strongly predicts electric range** ($r = -0.61$) — the oldest vehicles are the shortest-range. As fleet turnover accelerates, Washington's average range will improve naturally without policy intervention.
11. **PHEVs dominate rural registrations** — Yakima, Spokane, and eastern WA counties show higher PHEV proportions, likely reflecting longer intercity drives where full-BEV range remains a concern. Rural charging infrastructure is the missing enabler.
12. **The "Growth wave" (2016–2020) produced 49% of all registrations** and is now aging into prime replacement years (Vehicle Age 6–10). A replacement demand surge is forecast for 2026–2030 — automakers and utilities should prepare for a fleet renewal cycle.

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12. Recommendations

#	Insight Addressed	Recommendation	Expected Impact
1	Geographic concentration; Snohomish/Pierce growth	Deploy Level 2 and DC fast charging infrastructure in Snohomish, Pierce, and Clark counties with a density target of 1 charger per 50 registered EVs	15–20% increase in EV registrations in these counties within 24 months
2	17.9% CAFV ineligibility; PHEV short-range gap	Extend CAFV incentive eligibility to PHEVs with ≥ 20 mi range through a separate "Transition Fuel" tier; this captures ~18,000 current registrations	Increases eligible fleet to 95%+, improving consumer confidence in purchasing PHEVs as a stepping stone
3	Rural high-friction utility complexity	Mandate pre-emptive multi-agency grid coordination MOUs in counties with Utility Complexity Score ≥ 2.5 before EV density exceeds 500 vehicles/county	Reduces future infrastructure coordination cost by est. 30–40% compared to reactive upgrades
4	Luxury-skewed adoption; mass-market gap	Create a $3,000 state point-of-sale rebate for mass-market BEVs priced under $40,000 (targeting LEAF, BOLT EV, Kia EV6 segment)	Model projects 12,000+ additional mass-market BEV registrations in Year 1
5	Fleet replacement wave approaching (2026–2030)	Engage fleet operators and dealers with trade-in incentive programs for vehicles with Model Year ≤ 2016 (Vehicle Age ≥ 10) to accelerate replacement with modern long-range BEVs	Reduces average fleet age by 2–3 years; lifts average fleet range by ~40 mi

13. Limitations and Next Steps

Data Limitations

- **64% missing range data** in the raw dataset means population-level range analysis is based on a subset. PHEVs are disproportionately missing, skewing BEV/PHEV range comparisons.
- **Administrative registry data** reflects registration location, not where vehicles are actually driven. A King County registered vehicle may regularly commute to

Snohomish — charging demand is undercounted outside urban cores.

- **No income or demographic data** is linked to registrations, making direct socio-economic equity analysis impossible without external data join.
- **Snapshot, not panel** — the dataset captures the current registry state; historical deregistrations and vehicle retirements are not tracked.

Method Limitations

- Non-parametric statistical tests were used due to non-normal range distributions — effect sizes are not directly comparable across groups.
- Market Segment classification is based on brand-level assignment, not MSRP, which may misclassify some entry-level luxury or top-trim mass-market vehicles.

Suggested Future Work

1. **Join with Census income data** (ACS) by census tract to produce equity-adjusted adoption maps
2. **Time-series forecasting** (ARIMA / Prophet) on monthly registration counts to project 2027–2030 adoption
3. **Charging infrastructure gap analysis** — overlay with publicly available AFDC charger location data
4. **Panel data tracking** — request historical DOL data to analyse deregistration and fleet turnover patterns
5. **Grid capacity modelling** — combine Utility Complexity Score with utility published capacity data for infrastructure stress-testing

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14. Contribution Matrix

Role	Name	GitHub Username
Project Lead	Omved Nagre	OmvedNagre
Data Lead	Om Mishra	Lalilal0
ETL Lead	Gogulamudi JayaDeep	Jayadeep007
Analysis Lead	Sushant	Sushantydv01
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